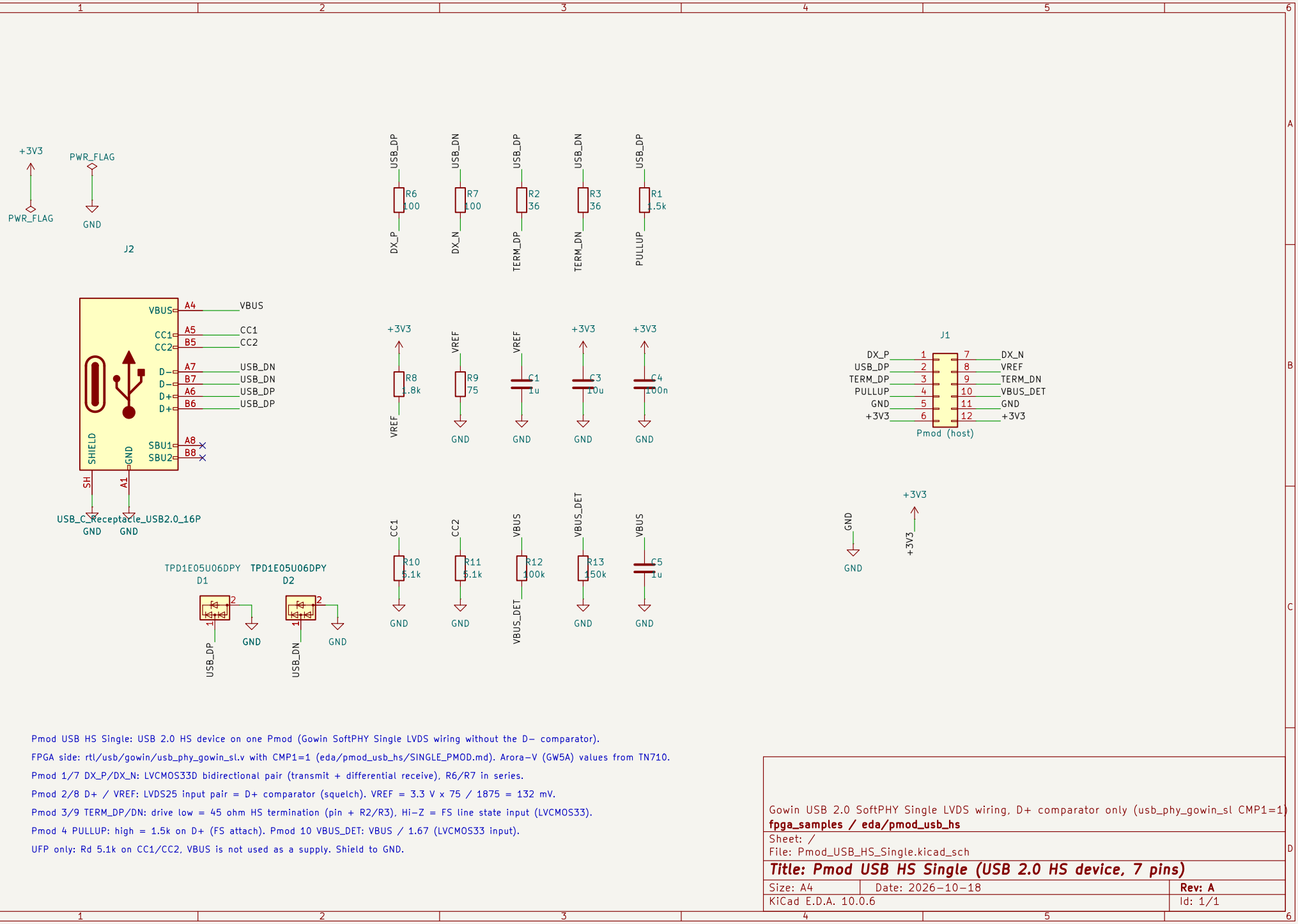


Pmod USB HS Single: USB 2.0 HS device on one Pmod (Gowin SoftPHY Single LVDS wiring without the D− comparator).

FPGA side: rtl/usb/gowin/usb_phy_gowin_sl.v with CMP1=1 (eda/pmod_usb_hs/SINGLE_PMOD.md). Arora−V (GW5A) values from TN710.

Pmod 1/7 DX_P/DX_N: LVCMOS33D bidirectional pair (transmit + differential receive), R6/R7 in series.

Pmod 2/8 D+ / VREF: LVDS25 input pair = D+ comparator (squelch). VREF = 3.3 V x 75 / 1875 = 132 mV.

Pmod 3/9 TERM_DP/DN: drive low = 45 ohm HS termination (pin + R2/R3), Hi−Z = FS line state input (LVCMOS33).

Pmod 4 PULLUP: high = 1.5k on D+ (FS attach). Pmod 10 VBUS_DET: VBUS / 1.67 (LVCMOS33 input).

UFP only: Rd 5.1k on CC1/CC2, VBUS is not used as a supply. Shield to GND.

Gowin USB 2.0 SoftPHY Single LVDS wiring, D+ comparator only (usb_phy_gowin_sl CMP1=1)

fpga_samples / eda/pmod_usb_hs

Sheet: /

File: Pmod_USB_HS_Single.kicad_sch

Title: Pmod USB HS Single (USB 2.0 HS device, 7 pins)

Size: A4	Date: 2026−10−18	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1